Tsinghua University 2025 Student Academic Research Promotion Program

Individual Project Mid-term Exchange

Preparation of analgesic peptide using synthetic biology approaches

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1. Background

Ziconotide is a peptide drug that provides effective pain relief without a risk of addiction. Its current use requires invasive intrathecal injection because it does not readily cross the blood-brain barrier.



Figure 1. Proposed colonization of engineered bacteria on the olfactory mucosal epithelium.

To develop a non-invasive alternative, we plan to engineer bacteria that colonize the nasal mucosa and secrete a Ziconotide fusion protein, bypassing the blood-brain barrier for non-invasive drug delivery.

2. Methods

Experimental workflow

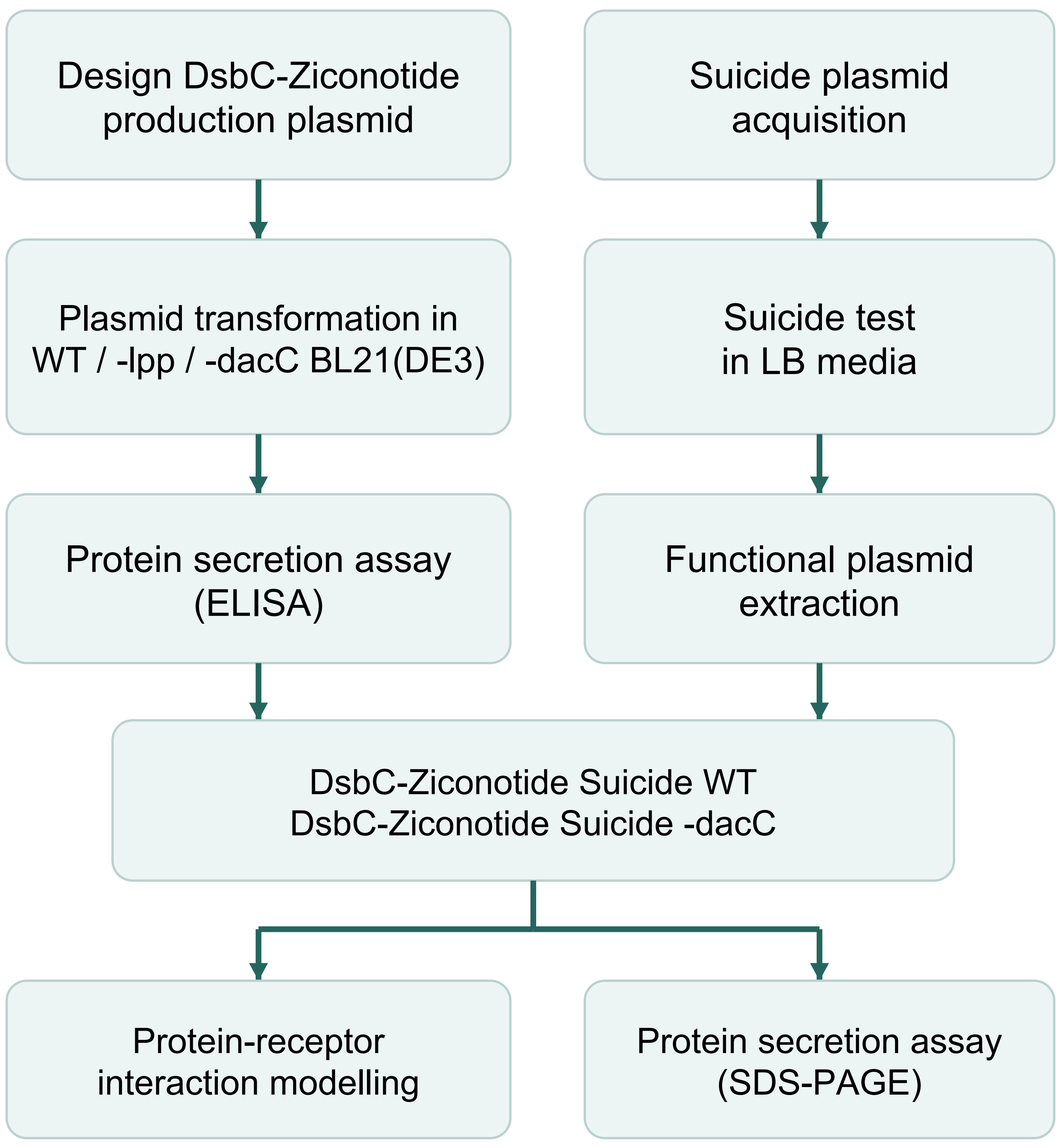


Figure 2. Overall experimental workflow. All steps have been completed in BL21(DE3) strains.

3. Results

Competitive ELISA

Absorbance in 2 x M9 medium

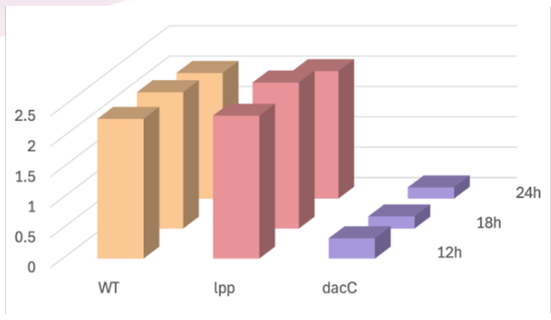


Figure 3. Competitive ELISA indicates that -dacC BL21(DE3) secretes the Ziconotide fusion protein.

Molecular dynamics

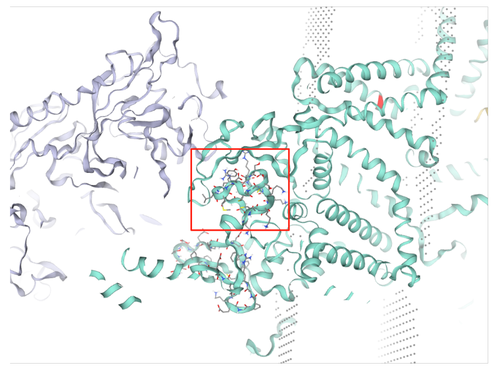


Figure 4. Molecular dynamics simulations indicate that the fusion protein can bind the Ziconotide receptor.

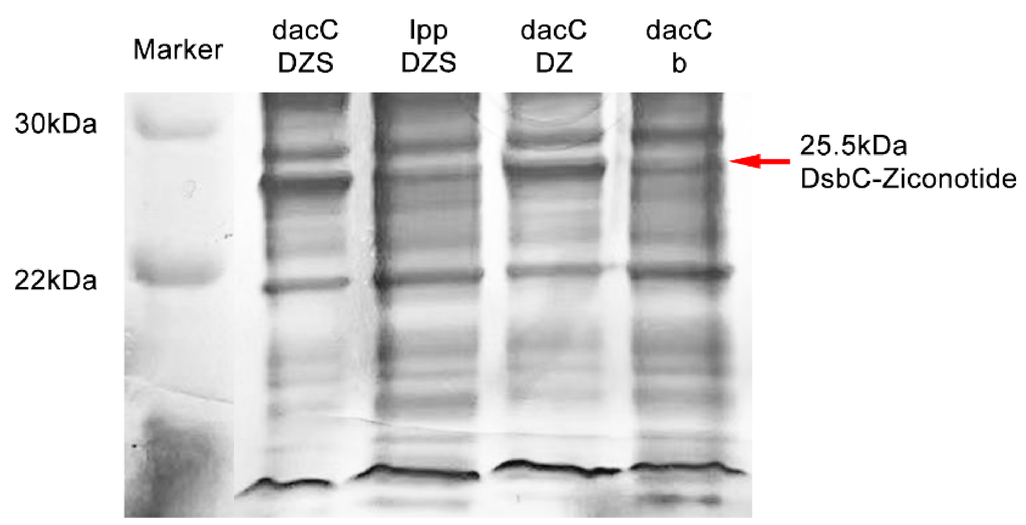


Figure 5. SDS-PAGE further supports successful secretion of the Ziconotide fusion protein.

4. Future Research Plan

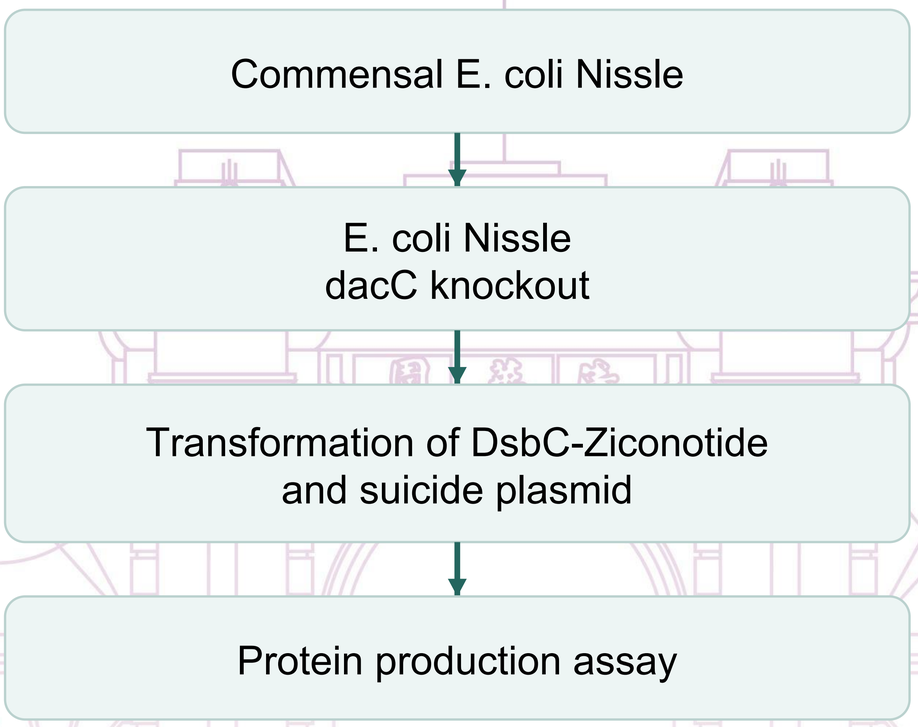


Figure 6. Planned protein secretion system in E. coli Nissle, a probiotic intended to stably colonize the nasal cavity.

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